

Lab Title:- HTTP ANALYSIS USING WIRESHARK AND TEXT TRAFFIC

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Course Name:- BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

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Date Of Submission:- 8/09/2026

Executive summary

The lab covers a practical hands on practical on some basic networking aspect required in digital forensics. These aspect include the creation of a simple webpage, capturing and analysis HTTP packet using wireshark(networking tools).

Lab Objectives

The objectives of the lab is to understand how to perform some basic http packet analysis using wireshark.

Tools and Resources Used

❖ **Kali linux:-** A penetration testing platform used for security assessments.

Kali linux tools

- Apache2:- this is the Apache HyperText Transfer Protocol (HTTP) server program that is designed to be run as a standalone daemon process.
 - wireshark:- is a GUI network protocol analyzer that lets you interactively browse packet data from a live network or from a previously saved capture file.
 - web browser(Mozilla browser):- a web browser for browsing/searching web pages.
 - sha256sum:- Print/compute or check SHA256 (256-bit) checksums message digest.
 - curl:- a tool for transferring data from or to a server using URLs.
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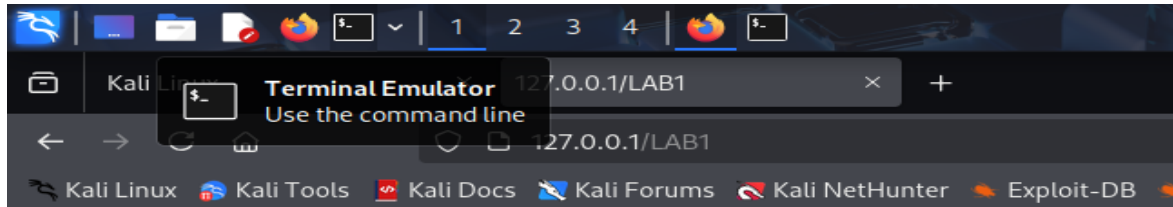
Methology

- I launched the kali linux terminal.
- I navigate to the html folder.
- I created a file(LAB1) in the html folder.
- I launched the Mozilla firefox browser.
- I search for the webpage created in the html folder(<http://127.0.0.1/LAB1>).

- I launched the wireshark to capture the LAB1 traffic from the Mozilla firefox browser.
- I analyse the LAB1 traffic.
- I save the LAB1 traffic as (NEWLAB1.pcapng) and then compute it sha256 hash value.

Screenshot and Evidence

SCREENSHOT SHOWING THE CREATED WEBPAGE



SBT-DF203 HTTP Evidence

Name: MUNTAKA MUHAMMAD BADAMASI

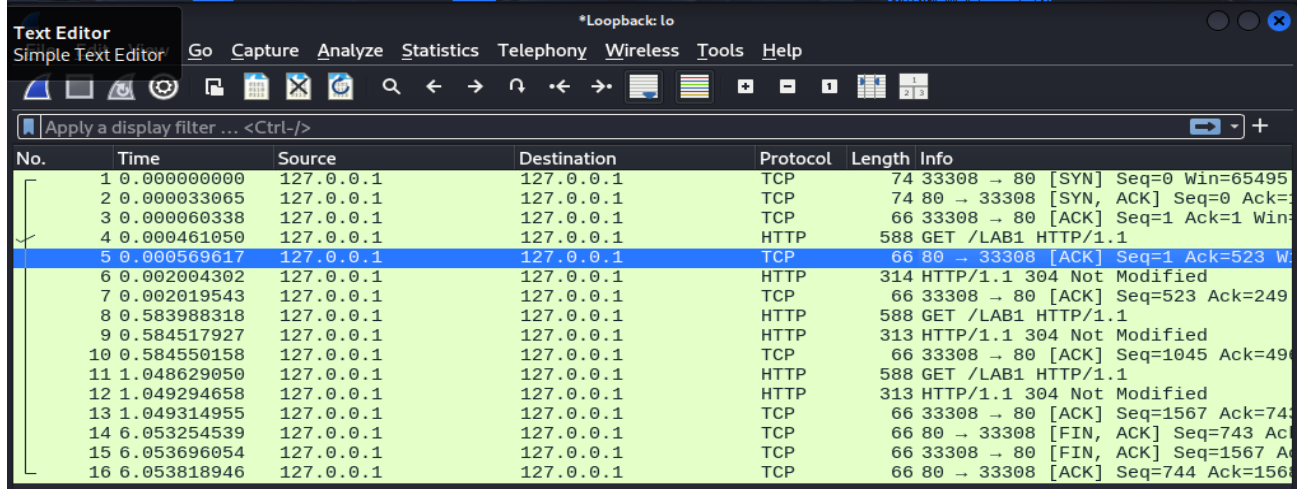
Reg No: 2025FWSD11563

SCREENSHOT SHOWING APACHE SERVICE LISTENING

```
File Actions Edit View Help
(root@kali)-[/]
# curl -v http://127.0.0.1/LAB1
* Trying 127.0.0.1:80...
* Connected to 127.0.0.1 (127.0.0.1) port 80
* using HTTP/1.x
> GET /LAB1 HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.12.1
> Accept: */*
< HTTP/1.1 200 OK
< Date: Mon, 07 Sep 2026 16:13:17 GMT
< Server: Apache/2.4.63 (Debian)
< Last-Modified: Mon, 07 Sep 2026 16:02:00 GMT
< ETag: "99-65ae6bd018d04"
< Accept-Ranges: bytes
< Content-Length: 153
<
<!DOCTYPE html>
<html>
<body>
<h1>SBT-DF203 HTTP Evidence</h1>
<p>Name: MUNTAKA MUHAMMAD BADAMASI</p>
<p>Reg No: 2025FWSD11563</p>
</body>
</html>
* Connection #0 to host 127.0.0.1 left intact
(root@kali)-[/]
```

SCREENSHOT SHOWING ACTIVE CAPTURE OF SELECTED INTERFACE

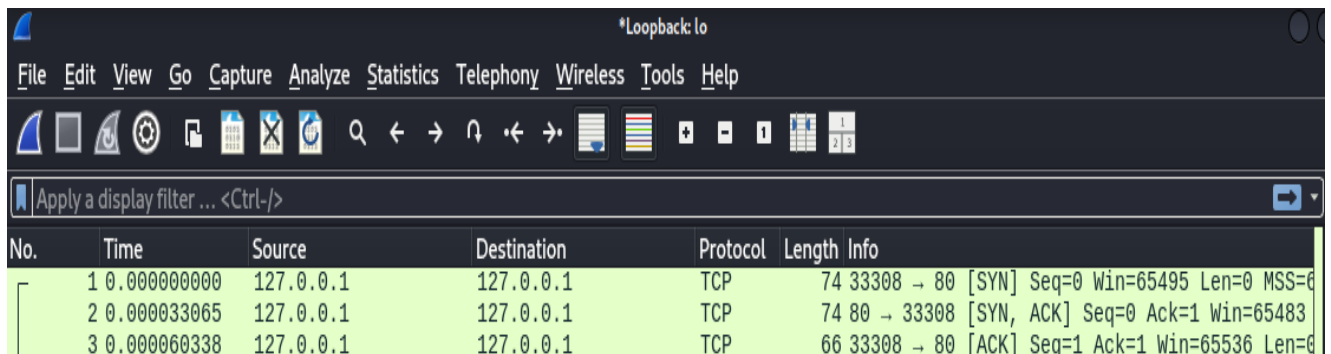
```
(root@kali)-[/home/kali/Desktop]
# wireshark -i lo
** (wireshark:26768) 12:42:18.348967 [Capture MESSAGE] -- Capture Start ...
** (wireshark:26768) 12:42:18.427237 [Capture MESSAGE] -- Capture started ...
** (wireshark:26768) 12:42:18.427356 [Capture MESSAGE] -- File: "/tmp/wireshark_Loopback0244U3.pcapng"
** (wireshark:26768) 12:42:30.420131 [Capture MESSAGE] -- Capture Stop ...
** (wireshark:26768) 12:42:30.452072 [Capture MESSAGE] -- Capture stopped...
```



The screenshot shows the Wireshark network protocol analyzer interface. The title bar indicates the capture is on the loopback interface '*Loopback: lo'. The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and analysis. A display filter is set to 'Apply a display filter ... <Ctrl-/>'. The packet list pane shows 16 captured packets. The selected packet (No. 5) is a TCP ACK from 127.0.0.1 to 127.0.0.1, sequence number 66, acknowledgment number 33308, and window size 523. The packet details pane shows the structure of the selected packet: Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	33308 → 80 [SYN] Seq=0 Win=65495
2	0.000033065	127.0.0.1	127.0.0.1	TCP	74	80 → 33308 [SYN, ACK] Seq=0 Ack=1
3	0.000060338	127.0.0.1	127.0.0.1	TCP	66	33308 → 80 [ACK] Seq=1 Ack=1 Win=
4	0.000461050	127.0.0.1	127.0.0.1	HTTP	588	GET /LAB1 HTTP/1.1
5	0.000569617	127.0.0.1	127.0.0.1	TCP	66	80 → 33308 [ACK] Seq=1 Ack=523 W
6	0.002004302	127.0.0.1	127.0.0.1	HTTP	314	HTTP/1.1 304 Not Modified
7	0.002019543	127.0.0.1	127.0.0.1	TCP	66	33308 → 80 [ACK] Seq=523 Ack=249
8	0.583988318	127.0.0.1	127.0.0.1	HTTP	588	GET /LAB1 HTTP/1.1
9	0.584517927	127.0.0.1	127.0.0.1	HTTP	313	HTTP/1.1 304 Not Modified
10	0.584550158	127.0.0.1	127.0.0.1	TCP	66	33308 → 80 [ACK] Seq=1045 Ack=49
11	1.048629050	127.0.0.1	127.0.0.1	HTTP	588	GET /LAB1 HTTP/1.1
12	1.049294658	127.0.0.1	127.0.0.1	HTTP	313	HTTP/1.1 304 Not Modified
13	1.049314955	127.0.0.1	127.0.0.1	TCP	66	33308 → 80 [ACK] Seq=1567 Ack=74
14	6.053254539	127.0.0.1	127.0.0.1	TCP	66	80 → 33308 [FIN, ACK] Seq=743 Ac
15	6.053696054	127.0.0.1	127.0.0.1	TCP	66	33308 → 80 [FIN, ACK] Seq=1567 A
16	6.053818946	127.0.0.1	127.0.0.1	TCP	66	80 → 33308 [ACK] Seq=744 Ack=156

SCREENSHOT SHOWING THREE WAY HANDSHAKE



The screenshot shows the Wireshark network protocol analyzer interface. The title bar indicates the capture is on the loopback interface '*Loopback: lo'. The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and analysis. A display filter is set to 'Apply a display filter ... <Ctrl-/>'. The packet list pane shows the first three packets of a TCP three-way handshake. The selected packet (No. 1) is a TCP SYN from 127.0.0.1 to 127.0.0.1, sequence number 0, window size 65495, and maximum segment size 65535.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	33308 → 80 [SYN] Seq=0 Win=65495 Len=0 MSS=6
2	0.000033065	127.0.0.1	127.0.0.1	TCP	74	80 → 33308 [SYN, ACK] Seq=0 Ack=1 Win=65483
3	0.000060338	127.0.0.1	127.0.0.1	TCP	66	33308 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0

SCREENSHOT SHOWING HTTP REQUEST

7 0.002019543	127.0.0.1	127.0.0.1	TCP	66 33308 → 80 [ACK] Seq=523 Ac
8 0.583988318	127.0.0.1	127.0.0.1	HTTP	588 GET /LAB1 HTTP/1.1
9 0.584517927	127.0.0.1	127.0.0.1	HTTP	313 HTTP/1.1 304 Not Modified

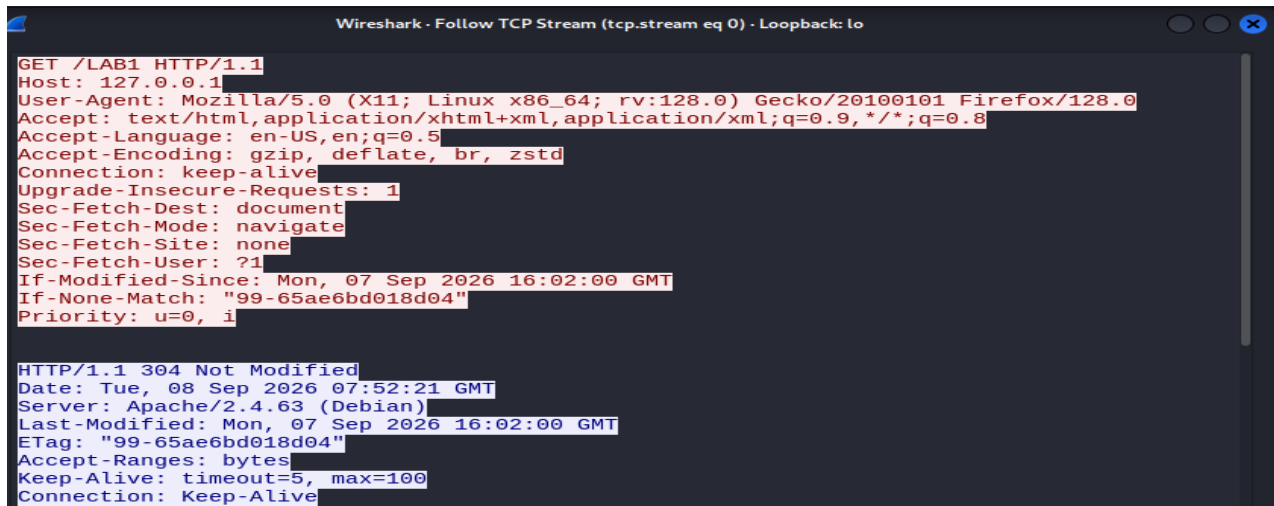
▶ Frame 9: 313 bytes on wire (2504 bits), 313 bytes captured (2504 bi ▶ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 ▶ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1 ▶ Transmission Control Protocol, Src Port: 80, Dst Port: 33308, Seq: ▼ Hypertext Transfer Protocol ▶ HTTP/1.1 304 Not Modified\r\n - Date: Tue, 08 Sep 2026 07:52:22 GMT\r\n - Server: Apache/2.4.63 (Debian)\r\n - Last-Modified: Mon, 07 Sep 2026 16:02:00 GMT\r\n - ETag: "99-65ae6bd018d04"\r\n - Accept-Ranges: bytes\r\n - Keep-Alive: timeout=5, max=99\r\n - Connection: Keep-Alive\r\n \r\n [Request in frame: 8]	<pre> 0000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 0010 01 2b 36 c0 40 00 40 06 05 0b 7f 6e 0a 00 00 00 0020 00 01 00 50 82 1c 10 3e f6 12 66 a3 00 00 00 00 0030 02 00 ff 1f 00 00 01 01 08 0a 66 2f 31 2e 31 20 3e 0040 f7 37 48 54 54 50 2f 31 2e 31 20 3e 00 00 00 00 0050 6f 74 20 4d 6f 64 69 66 69 65 64 63 00 00 00 00 0060 65 3a 20 54 75 65 2c 20 30 38 20 3e 00 00 00 00 0070 30 32 36 20 30 37 3a 35 32 3a 32 3e 00 00 00 00 0080 0d 0a 53 65 72 76 65 72 3a 20 41 7f 6e 0a 00 00 0090 2f 32 2e 34 2e 36 33 20 28 44 65 63 00 00 00 00 00a0 0d 0a 4c 61 73 74 2d 4d 6f 64 69 63 00 00 00 00 00b0 20 4d 6f 6e 2c 20 30 37 20 53 65 7f 6e 0a 00 00 00c0 36 20 31 36 3a 30 32 3a 30 30 20 4d 6f 64 69 63 00d0 45 54 61 67 3a 20 22 39 39 2d 36 63 00 00 00 00 00e0 64 30 31 38 64 30 34 22 0d 0a 41 7f 6e 0a 00 00 </pre>
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SCREENSHOT SHOWING HTTP 200 RESPONSE

6 0.002004302	127.0.0.1	127.0.0.1	HTTP	314 HTTP/1.1 304 Not Modified
7 0.002019543	127.0.0.1	127.0.0.1	TCP	66 33308 → 80 [ACK] Seq=523 Ac
8 0.583988318	127.0.0.1	127.0.0.1	HTTP	588 GET /LAB1 HTTP/1.1
9 0.584517927	127.0.0.1	127.0.0.1	HTTP	313 HTTP/1.1 304 Not Modified

▶ Frame 8: 588 bytes on wire (4704 bits), 588 bytes captured (4704 bi ▶ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 ▶ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1 ▶ Transmission Control Protocol, Src Port: 33308, Dst Port: 80, Seq: ▼ Hypertext Transfer Protocol ▶ GET /LAB1 HTTP/1.1\r\n - Host: 127.0.0.1\r\n - User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/* - Accept-Language: en-US,en;q=0.5\r\n - Accept-Encoding: gzip, deflate, br, zstd\r\n - Connection: keep-alive\r\n - Upgrade-Insecure-Requests: 1\r\n - Sec-Fetch-Dest: document\r\n - Sec-Fetch-Mode: navigate\r\n - Sec-Fetch-Dest: document\r\n - Sec-Fetch-Mode: navigate\r\n - Sec-Fetch-Site: none\r\n - Sec-Fetch-User: ?1\r\n - If-Modified-Since: Mon, 07 Sep 2026 16:02:00 GMT\r\n - If-None-Match: "99-65ae6bd018d04"\r\n - Priority: u=0, i\r\n \r\n [Response in frame: 9] [Full request URI: http://127.0.0.1/LAB1]	<pre> 0000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 0010 02 3e 12 d7 40 00 40 06 27 e1 7f 6e 0a 00 00 00 0020 00 01 82 1c 00 50 66 ac af 09 10 00 00 00 00 00 0030 02 00 00 33 00 00 01 01 08 0a 66 2f 31 2e 31 20 3e 0040 f4 f1 47 45 54 20 2f 4c 41 42 31 2e 31 20 3e 00 0050 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 54 75 65 2c 0060 30 2e 30 2e 31 0d 0a 55 73 65 72 30 32 36 20 30 0070 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 58 31 31 3b 20 0080 4c 69 6e 75 78 20 34 3b 20 72 76 3a 31 32 38 2e 0090 30 6b 6f 2f 32 30 31 30 30 31 30 31 6b 6f 2f 32 00a0 30 31 30 30 66 6f 78 2f 31 32 38 2e 30 0d 0a 74 00b0 3a 20 74 65 78 74 2f 68 74 6d 6c 69 63 61 74 00c0 69 6f 6e 2f 78 68 6d 6c 6c 2c 61 70 70 6c 69 00d0 63 61 74 6d 6c 2c 61 70 70 6c 69 63 61 74 00e0 6d 6c 2c 61 70 70 6c 69 63 61 74 6d 6c 2c 61 0100 70 70 6c 69 63 61 74 6d 6c 2c 61 70 70 6c 69 0110 63 61 74 6d 6c 2c 61 70 70 6c 69 63 61 74 6d 0120 6c 2c 61 70 70 6c 69 63 61 74 6d 6c 2c 61 70 0130 70 6c 69 63 61 74 6d 6c 2c 61 70 70 6c 69 63 0140 61 74 65 2c 20 62 72 2c 20 7a 73 6f 6e 6e 65 0150 63 74 69 6f 6e 3a 20 61 6c 69 76 65 0d 0a 55 0160 70 67 72 61 70 65 69 75 70 65 64 50 65 74 </pre>
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SCREENSHOT SHOWING TCP STREAM

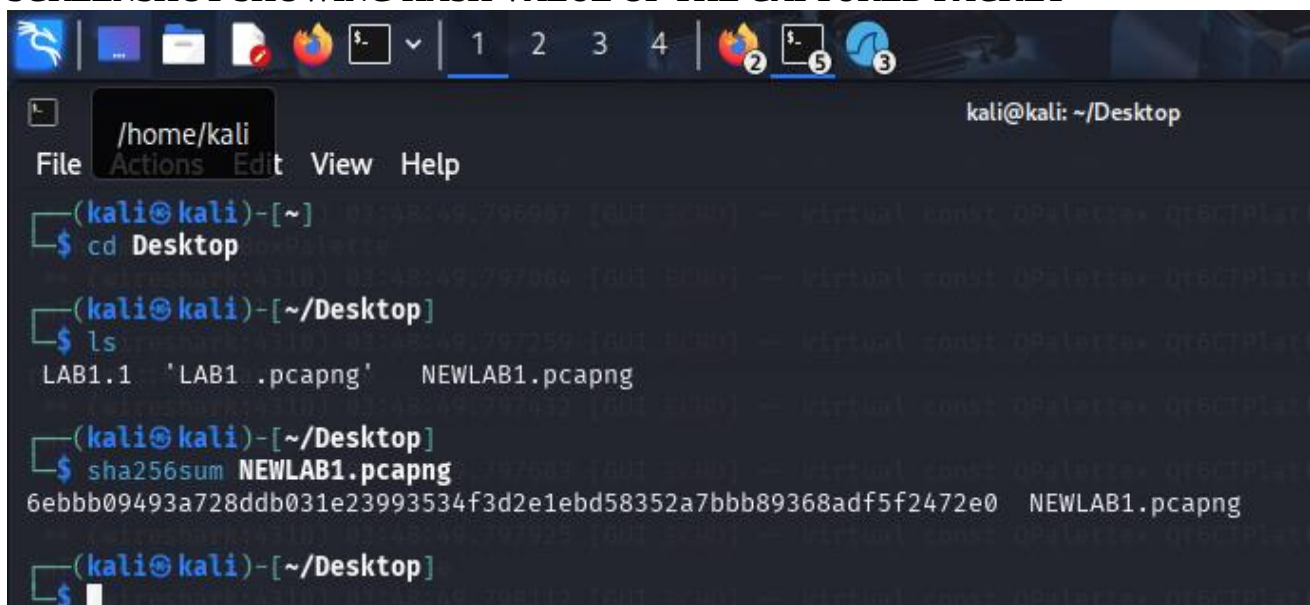


The screenshot shows the 'Follow TCP Stream' window in Wireshark, displaying the raw data of a captured packet. The data is divided into two sections: the request and the response. The request is an HTTP GET for '/LAB1' from 127.0.0.1 using Mozilla/5.0 as the user-agent. The response is an HTTP 304 'Not Modified' status from an Apache server, indicating the resource has not changed since the last request.

```
GET /LAB1 HTTP/1.1
Host: 127.0.0.1
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br, zstd
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Sec-Fetch-Dest: document
Sec-Fetch-Mode: navigate
Sec-Fetch-Site: none
Sec-Fetch-User: ?1
If-Modified-Since: Mon, 07 Sep 2026 16:02:00 GMT
If-None-Match: "99-65ae6bd018d04"
Priority: u=0, i

HTTP/1.1 304 Not Modified
Date: Tue, 08 Sep 2026 07:52:21 GMT
Server: Apache/2.4.63 (Debian)
Last-Modified: Mon, 07 Sep 2026 16:02:00 GMT
ETag: "99-65ae6bd018d04"
Accept-Ranges: bytes
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
```

SCREENSHOT SHOWING HASH VALUE OF THE CAPTURED PACKET



The screenshot shows a terminal window on a Kali Linux system. The user navigates to the Desktop directory and lists files, finding 'LAB1.pcapng' and 'NEWLAB1.pcapng'. They then calculate the SHA256 hash of 'NEWLAB1.pcapng', resulting in a long hexadecimal string.

```
kali@kali: ~/Desktop
File Actions Edit View Help
(kali@kali)-[~]
$ cd Desktop
(kali@kali)-[~/Desktop]
$ ls
LAB1.1 'LAB1.pcapng' NEWLAB1.pcapng
(kali@kali)-[~/Desktop]
$ sha256sum NEWLAB1.pcapng
6ebbb09493a728ddb031e23993534f3d2e1ebd58352a7bbb89368adf5f2472e0 NEWLAB1.pcapng
(kali@kali)-[~/Desktop]
$
```

Conclusion

In conclusion this lab has help me in understanding some of the practical aspect behind three-way handshake and also how to analyse a captured network traffic to identify some information such as the source/destination ip address/port number, HTTP request/response frame, user-agent, host etc using a very important networking tool called the wireshark.

Reference

➤ICDFA course Material (SBT-DF203(basic networking skills for digital forensics)- by
COMMANDANT AMINU IDRIS

➤ICDFA Recorded sessions
